Multi-Epitope Vaccine (MEV) Design Report

Protein Metadata

Job ID:	30a297d1-2455-43f3-a16e-28eaff28fd42
UniProt ID:	P56856-2
Status:	running
Created:	2026-03-14T18:08:10.694447
Completed:	None
Protein Name:	Isoform A2 of Claudin-18
Organism:	Homo sapiens
Sequence Length:	261 AA

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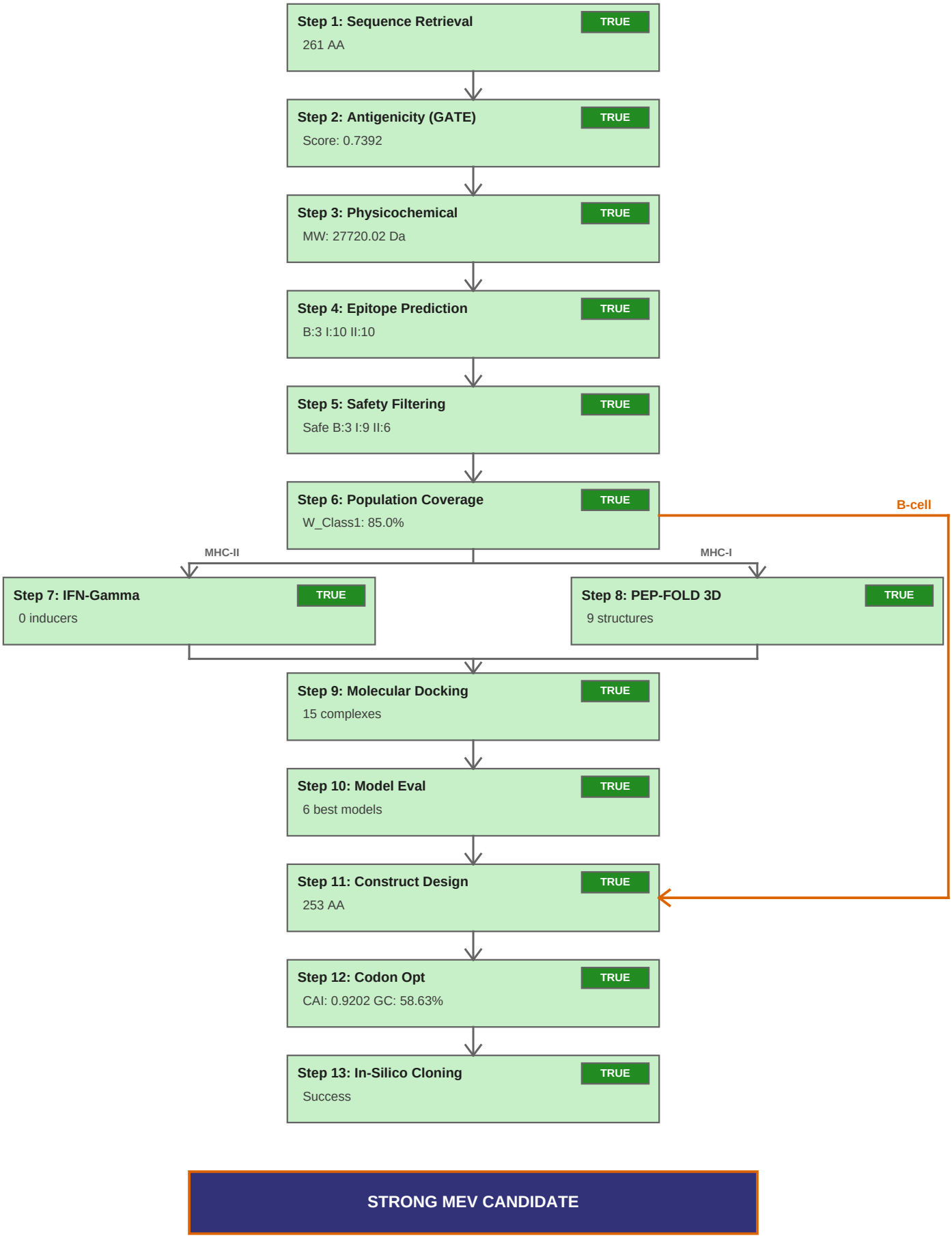
Pipeline Execution Flowchart



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Step 01 - Sequence Retrieval [PASSED]

uniprot_id:	P56856-2
isoform:	True
sequence:	MAVTACQGLGFVVSLIGIAGIIAATCMDQWSTQDLYNNPVTAVFNYQGLWRSCVRES SGFTECRGYFTLLGLPAMLQAVRALMIVGIVLGAIGLLVSIFALKCIRIGSMEDSAKANM TLTSGIMFIVSGLCAIAGVSVFANMLVTNFWMSTANMYTGMGGMVQTVQTRYTFGAA LFVGWVAGGLTLIGGVMMCIACRGLAPEETNYKAVSYHASGHSVAYKPGGFKASTGF GSNTKNKKIYDGGARTEDEVQSYPSKHDYV
length:	261
protein_name:	Isoform A2 of Claudin-18
organism:	Homo sapiens

Step 02 - Antigenicity Screening [PASSED]

antigenicity_score:	0.7392
classification:	Antigen
gate_passed:	True
tool:	Kolaskar & Tongaonkar (1990) [fallback]
threshold:	0.4

Step 03 - Physicochemical Analysis [PASSED]

molecular_weight:	27720.02
theoretical_pi:	8.6
instability_index:	32.05
aliphatic_index:	89.69
gravy:	0.448

Step 04 - Epitope Prediction [PASSED]

bcell_epitopes:	(complex - see JSON report)
bcell_count:	3
mhc1_epitopes:	(complex - see JSON report)
mhc1_count:	10
mhc2_epitopes:	(complex - see JSON report)
mhc2_count:	10
tool:	IEDB (Bepipred-2.0 / NetMHCpan-4.1 / NetMHCIIpan-4.0)

Step 05 - Safety Filtering [PASSED]

safe_bcell_epitopes:	(complex - see JSON report)
safe_bcell_count:	3
safe_mhc1_epitopes:	(complex - see JSON report)
safe_mhc1_count:	9
safe_mhc2_epitopes:	(complex - see JSON report)
safe_mhc2_count:	6

bcell_removed: 0

mhc1_removed: 1

mhc2_removed: 4

filtering_criteria: [
"toxic",
"allergenic",
"non-antigenic"
]

Step 06 - Population Coverage [PASSED]

bcell_peptides_used: [

"VFANMLVTNFWMSTANMYTGMGGMVQTVQTRYTFGAALFVGWVAGGLTLIGGVM
MCIACRGLAPEETNYKAVSYHASGHSVAYKPGGFKASTGFGSNTKNKKIYDGGARTE
DEVQSYPSKHDYV",

"VTACQGLGFVVSLIGIAGIIAATCMDQWSTQDLYNNPVTAVFNYQGLWRSCVRESSG
FTECRGYFTLLGLPAMLQAVRA",
"GAIGLLVSIFALKCIRIGSMEDSAKANMTLTSG"
]

mhc1_alleles_used: [
"HLA-A*26:01",
"HLA-C*07:01",
"HLA-A*23:01",
"HLA-A*24:02",
"HLA-A*29:02",
"HLA-A*32:01",
"HLA-A*11:01",
"HLA-B*35:01"
]

mhc2_alleles_used: [
"HLA-DPA1*01:03/DPB1*02:01",
"HLA-DPA1*02:01/DPB1*01:01",
"HLA-DRB3*02:02"
]

bcell_population_coverage: {
"North America": 0.98,
"Europe": 0.96,
"East Asia": 0.91,
"South Asia": 0.94,
"Africa": 0.87,
"World": 0.93
}

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class1_population_coverage:

{
"North America": 0.9,
"Europe": 0.88,
"East Asia": 0.83,
"South Asia": 0.86,
"Africa": 0.79,
"World": 0.85
}

class2_population_coverage:

{
"North America": 0.86,
"Europe": 0.84,
"East Asia": 0.79,
"South Asia": 0.82,
"Africa": 0.75,
"World": 0.81
}

world_bcell_coverage_pct: 93.0

world_class1_coverage_pct: 85.0

world_class2_coverage_pct: 81.0

Step 07 - IFN-Gamma Prediction [PASSED]

approach: Motif and SVM hybrid

model: IFN-gamma versus Non IFN-gamma

ifn_gamma_epitopes: []

ifn_gamma_count: 0

errors: []

Step 08 - PEP-FOLD Modeling [PASSED]

peptides_submitted: 9

successful_models: 9

pepfold_models: (complex - see JSON report)

skipped_invalid_length: []

errors: []

Step 09 - Molecular Docking [PASSED]

docked_complexes: (complex - see JSON report)

total_docked: 15

Step 10 - Model Evaluation [PASSED]

evaluated_complexes: (complex - see JSON report)

best_models: (complex - see JSON report)

best_model_count: 6

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Step 11 - Construct Design [PASSED]

adjuvant_n_terminal: APPHALS

adjuvant_c_terminal: KDPNAPKRPPSAFFLFCSE

padre_sequence: AAATLWKAAKFVA

mhc2_epitopes_used:
[
"TQDLYNNPVTAVFNY",
"ETNYKAVSYHASGHS"
]

mhc1_epitopes_used:
[
"RYTFGAALF",
"ETNYKAVSY",
"HASGHSVAY",
"ASGHSVAYK"
]

bcell_epitopes_used:
[

"VFANMLVTNFWMSTANMYTGMGGMVQTVQTRYTFGAALFVGWVAGGLTLIGGVM
MCIACRGLAPEETNYKAVSYHASGHSVAYKPGGFKASTGFGSNTKNKKIYDGGARTE
DEVQSYPSKHDYV"
]

final_construct_sequence: APPHALSEAAAKAAATLWKAAKFVATQDLYNNPVTAVFNYGPGPGETNYKAVSYHAS
GHSRYTFGAALFAAYETNYKAVSYAAYHASGHSVAYAAYASGHSVAYKVFANMLVTN
FWMSTANMYTGMGGMVQTVQTRYTFGAALFVGWVAGGLTLIGGVMMCIACRGLAP
EETNYKAVSYHASGHSVAYKPGGFKASTGFGSNTKNKKIYDGGARTEDEVQSYPSKH
DYVEAAAKKDPNAPKRPPSAFFLFCSE

construct_length_aa: 253

Step 12 - Codon Optimization [PASSED]

expression_host: E. coli

optimized_dna_sequence: (759 chars - see JSON report)

dna_length_bp: 759

cai_score: 0.9202

gc_content_pct: 58.63

Step 13 - In-Silico Cloning [PASSED]

vector: pShuttle-CMV

insert_length_bp: 759

orientation: Forward (5' CMV promoter ? Insert ? 3' BGH polyA)

restriction_site_conflicts: []

cloning_compatible: True

cloning_status: Success

cloning_map: {

"vector": "pShuttle-CMV",
"insert_size_bp": 759,
"cloning_sites": {
"upstream": "NotI (5')",
"downstream": "XhoI (3')"
},
"promoter": "CMV immediate-early",
"terminator": "BGH poly-A signal",
"orientation": "Forward (5' CMV promoter \u2192 Insert \u2192 3' BGH polyA)"
}

Overall MEV Candidacy Assessment

STRONG MEV CANDIDATE